

Toward 1% single-photon anharmonicity with periodically poled lithium niobate microring resonators

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The absence of the single-photon nonlinearity has been a major roadblock in developing quantum photonic circuits at optical frequencies. In this paper, we demonstrate a periodically poled thin film lithium niobate microring resonator (PPLNMR) that reaches 5,000,000%/W second-harmonic conversion efficiency—almost 20-fold enhancement over the state-of-the-art—by accessing its largest $\chi^{(2)}$ tensor component d_{33} via quasi-phase matching. The corresponding single-photon coupling rate $g/2\pi$ is estimated to be 1.2 MHz, which is an important milestone as it approaches the dissipation rate $\kappa/2\pi$ of best-available lithium niobate microresonators developed in the community. Using a figure of merit defined as g/κ , our device reaches a single-photon nonlinear anharmonicity approaching 1%. We show that, by further scaling of the device, it is possible to improve the single-photon anharmonicity to a regime where photon blockade effect can be manifested. © 2020 Optical Society of America under the terms of the OSA Open Access Publishing Agreement

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1. INTRODUCTION

Quantum photonic integrated circuits have received growing interests since such platforms offer the stability and integrability toward solid-state quantum applications [1–6]. By encoding quantum information into the optical photons, the quantum information processing, quantum communication, and quantum metrology would benefit from the merits of the bosonic carriers, including the high propagation velocity, long propagation distance, and infinite-dimensional Hilbert spaces. Microwave photons could be processed by superconducting quantum circuits, where high-fidelity quantum operations approaching error-correction thresholds are achieved by the lossless nonlinearity inherent to Josephson effect [7,8]. However, at optical frequencies, the absence of the single-photon nonlinearity hinders the quest for the deterministic single-photon sources and further high-fidelity photonic quantum gates, which are two main building blocks for the scalable quantum circuits [9–13]. While the photon–photon interactions required to realize quantum gate can be mediated through light–matter interaction with atomic or atom-like solid-state emitters, routes to scaling such systems remain challenging [14–16].

Leveraging the small mode volume and enhanced photon–photon interaction in high quality factor (Q) optical microcavities [17], as well as the inherent material nonlinearity [18], several schemes based on the microcavity made by the materials with second-order ($\chi^{(2)}$) nonlinearity have been theoretically proposed to allow the indistinguishable single-photon state, as well as the

deterministic photon–photon gates via photon blockade effect by harnessing the single-photon nonlinear anharmonicity [19–23] and thereby bringing in-sight scalable quantum photonic computing. Lithium niobate (LN) has recently risen to the forefront of integrated quantum photonics circuits due to its intrinsic strong optical nonlinearity, electro-optic effect, and experimentally demonstrated ultra-low-loss nanophotonics platform for scaling. In this paper, we describe a 20-fold enhancement over state-of-the-art devices in second-harmonic generation (SHG) [24,25] with the thin film periodically poled LN microring resonators (PPLNMRs) by leveraging its largest $\chi^{(2)}$ tensor element d_{33} for quasi-phase matching. This exceptionally high nonlinearity translates to a vacuum photon–photon coupling strength $g/2\pi$ of 1.2 MHz, which is an important milestone as it approaches the dissipation rate κ of best-available LN microresonators [26]. Currently, the trade-off between mode confinement and optical loss limits our device to a figure of merit (FOM) single-photon anharmonicity, defined as g/κ , at 10^{-2} . Upon further scaling of microresonators, it is possible to improve this FOM by an order of magnitude, making it feasible to realize the photon blockade effect in certain specifically designed device configuration.

Single-photon anharmonicity is appealing for quantum photonics applications. As an example, Fig. 1 illustrates a conceptual photon blockade device leveraging single-photon-level $\chi^{(2)}$ nonlinearity based on a PPLNMR, which could generate single photons with sub-Poissonian quantum statistics from a classical laser input. Photons with carrier frequency ω_a (blue) travel in a

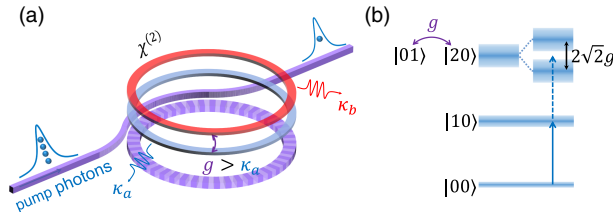


Fig. 1. (a) Photon blockade effect due to $\chi^{(2)}$ single-photon nonlinear anharmonicity in a doubly resonant cavity based on PPLNMR. (b) The schematic energy level diagram of a PPLNMR with single-photon nonlinearity. The anharmonicity of the nonlinear system is determined by the energy level splitting relative to their widths.

waveguide and couple to the cavity mode a with a dissipation rate κ_a . With the fulfillment of certain phase-matching conditions for the degenerate three-wave mixing, a significant nonlinear coupling strength g between mode a and mode b (with a frequency of $\omega_b = 2\omega_a$) could be obtained, and thereby a frequency conversion from ω_a to ω_b is feasible. For example, the highly efficient SHG between mode a in the telecom band and mode b in the near-visible band has been experimentally demonstrated in a PPLNMR [24]. The system Hamiltonian of such PPLNMR device reads

$$\hat{H}/\hbar = \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} + g((\hat{a}^\dagger)^2 \hat{b} + \hat{a}^2 \hat{b}^\dagger). \quad (1)$$

Here, $\hat{a}$ and $\hat{b}$ represent the bosonic operators, and g denotes the vacuum nonlinear photon–photon interaction strength between mode a and b . Considering only a few excitations, the system energy levels can be written as $|mn\rangle = |m\rangle_a \otimes |n\rangle_b$ in the Fock state basis, with $m, n \in \mathbb{Z}$. When the single-photon nonlinearity is realized, i.e., the coupling strength exceeds the dissipation

rate, the state $|20\rangle$ strongly couples to $|01\rangle$, and new eigenstates $(|20\rangle \pm |01\rangle)/\sqrt{2}$ with a frequency splitting of $2\sqrt{2}g$ are produced [Fig. 1(b)]. The induced anharmonicity of energy levels gives rise to the photon blockade effect as illustrated in Fig. 1(a), where the pump mode energy level with photon number $N \geq 2$ is no longer resonant with the cavity, and thereby the two-excitation state would be inhibited if the detuning $\sqrt{2}g$ is comparable or larger than the energy linewidth κ_a , as indicated by the blue dashed arrow in Fig. 1(b). Hence, we introduce the dimensionless FOM quantifying the anharmonicity of the intracavity Fock states at the single-photon level, $\text{FOM} = g/\kappa_a$. The larger the FOM is, the better performance it has in single-photon generation. Alternatively, the FOM also indicates the number of quantum gate operations on single photons before significant fidelity loss due to the photon dissipation to environment.

2. OPTIMIZATION OF COUPLING STRENGTH g

Based on the above discussion, a larger coupling strength g is always demanding to realize single-photon anharmonicity, since the mode dissipation rates are mostly restricted by the material and fabrication technique in practices. In a microring resonator, the photon–photon coupling strength g is determined by the material $\chi^{(2)}$ coefficient, modal overlap factor γ , and the mode volume V through the relation $g \propto \chi^{(2)}\gamma/\sqrt{V}$. According to Ref. [24], the PPLNMR is employed for high $\chi^{(2)}$ coefficient and phase-matching condition, and a SHG efficiency of 250,000% was achieved. Here, a significant improvement of g is demonstrated by solving two practical challenges: the utilization of largest nonlinear term d_{33} of the z-cut LN thin film and the high-fidelity radial poling of the microring.

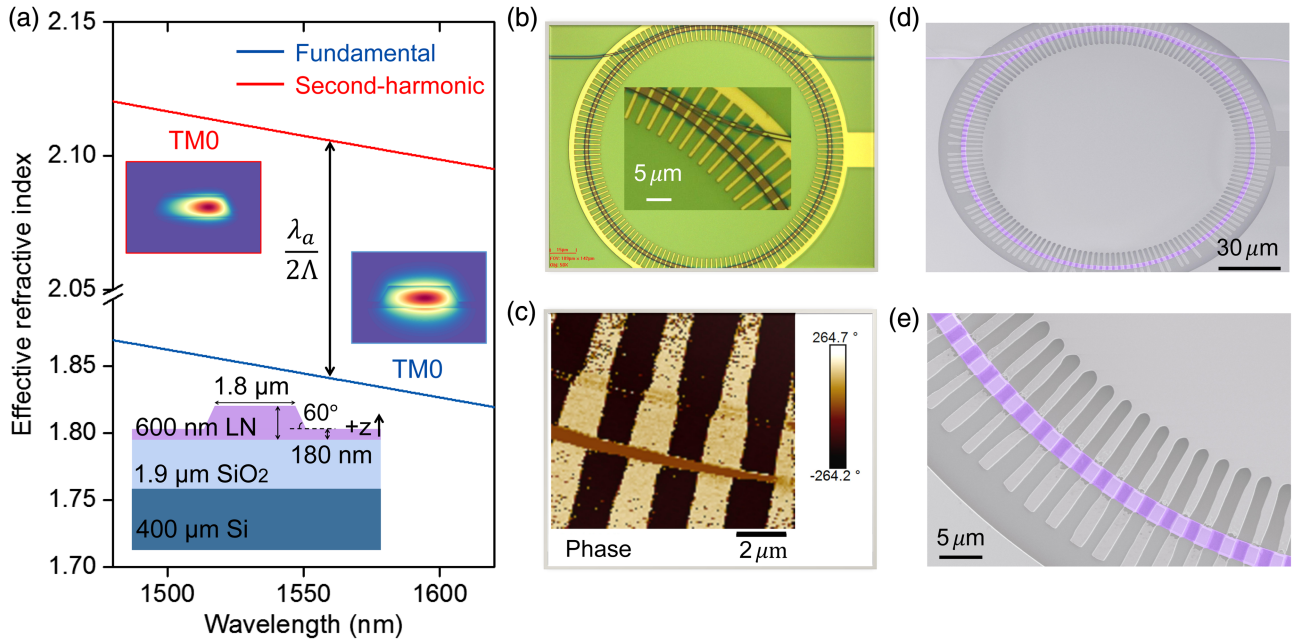


Fig. 2. (a) Effective refractive indices of the TM fundamental and SH modes, where the criterion for the type-0, first-order QPM between mode a at 1560 nm and mode b at 780 nm with a poling period $\Lambda = 2.95 \mu\text{m}$ is indicated by the vertical, double-headed arrow. The upper insets show the simulated profiles (amplitude of the vertical electric-field component) of the mode a and b while the lower inset depicts the schematic cross section of a z-cut PPLNMR with a radius of $70 \mu\text{m}$. (b) Optical image of the etched microring with radial poling electrodes and its zoomed view around the waveguide-microring coupling region. (c) PFM phase scan over a small portion of a PPLNMR, revealing the alternate ferroelectric domain structures and a duty cycle close to 50%. (d) False-color SEM image of a duplicate PPLNMR etched in HF and its zoomed view (e) reveal a high-fidelity periodic poling along the whole microring. Dark purple, inverted domains; light purple, uninverted domains.

First, since the optic axis of z-cut LN lies vertically, to employ its d_{33} term, we design for phase matching between the fundamental quasi-transverse magnetic (TM) mode a at 1560 nm and second-harmonic (SH) mode b at 780 nm, as shown in Fig. 2(a). The lower inset depicts the schematic cross section of a partially etched z-cut LN microring with a radius of 70 μm . The fabrication of the z-cut air-cladded LN microrings is detailed in Ref. [27]. The lowest-order SH mode is favorable due to the lower scattering loss and larger modal overlap factor. The simulated profiles (amplitude of the vertical electric-field component) for the fundamental TM0 and SH TM0 modes are presented in the upper insets of Fig. 2. Due to their large refractive index difference, the momentum conservation could only be satisfied via quasi-phase matching with a poling period of $\Lambda = \lambda_a/2(n_b - n_a) = 2.95 \mu\text{m}$ as indicated by the double-headed black arrow in Fig. 2(a). Consequently, an optimized $g/2\pi$ is calculated to be 1.78 MHz according to Eq. (2) in Ref. [24].

Second, for the implementation of quasi-phase matching, the poling electrodes with a tooth width of 750 nm are deposited on top of the etched LN microring, as shown in Fig. 2(b). The tooth width is designed to be smaller than $\Lambda/2$ to allow for the inevitable lateral domain broadening and ensure a duty cycle of $\sim 50\%$. The periodic ferroelectric domain inversion is then enabled by keeping the silicon substrate as the electrical ground while applying several 600 V, 250 ms pulses at an elevated temperature, as elaborated in Ref. [24]. After removing the electrodes, piezoresponse force microscopy (PFM) is utilized as a non-destructive way to visualize the alternate domain inversion as presented in Fig. 2(c), where the dark regions correspond to the inverted domains and a duty cycle close to 50% is achieved. Moreover, the difference in etch rates

between the poled and unpoled regions of z-cut LN in hydrofluoric acid (HF) allows us to examine the poling quality along the whole microring under a scanning electron microscope (SEM). Figures 2(d) and 2(e) are the false-color SEM images of a PPLNMR mock-up etched in HF, which implies a high-fidelity periodic poling along the microring. A period of $\sim 2.95 \mu\text{m}$ and a duty cycle of $\sim 50\%$ are confirmed.

3. CHARACTERIZATION OF g

The SHG measurement is implemented to verify the nonlinear coupling strength g of the fabricated PPLNMR device. With a pump field near the fundamental frequency, the system can be described by the Hamiltonian,

$$\hat{H}/\hbar = \omega_a \hat{a}^\dagger \hat{a} + \omega_b \hat{b}^\dagger \hat{b} + g((\hat{a}^\dagger)^2 \hat{b} + \hat{a}^2 \hat{b}^\dagger) + i\epsilon_p(-\hat{a}e^{i\omega_p t} + \hat{a}^\dagger e^{-i\omega_p t}), \quad (2)$$

where $\epsilon_p = \sqrt{2\kappa_{a,1} P_{a,\text{in}}/\hbar\omega_p}$ is the input pump strength, $P_{a,\text{in}}$ is the on-chip pump power, and $\kappa_{a,1}$ denotes the external dissipation rate of the mode a . At the steady state, the on-chip transmission of the pump laser and the output power of the SH signal are derived as

$$P_{a,\text{out}} = \frac{\delta_a^2 + (\kappa_{a,0} - \kappa_{a,1})^2}{\delta_a^2 + \kappa_a^2} P_{a,\text{in}}, \quad (3)$$

$$P_{b,\text{out}} = \frac{g^2 P_{a,\text{in}}^2 \omega_b}{\hbar \omega_p^2} \frac{2\kappa_{b,1}}{\delta_b^2 + \kappa_b^2} \left(\frac{2\kappa_{a,1}}{\delta_a^2 + \kappa_a^2} \right)^2, \quad (4)$$

where κ_1 , κ_0 , and κ are, respectively, the external, intrinsic, and total dissipation rates of the cavity mode with $\kappa = \kappa_1 + \kappa_0$.

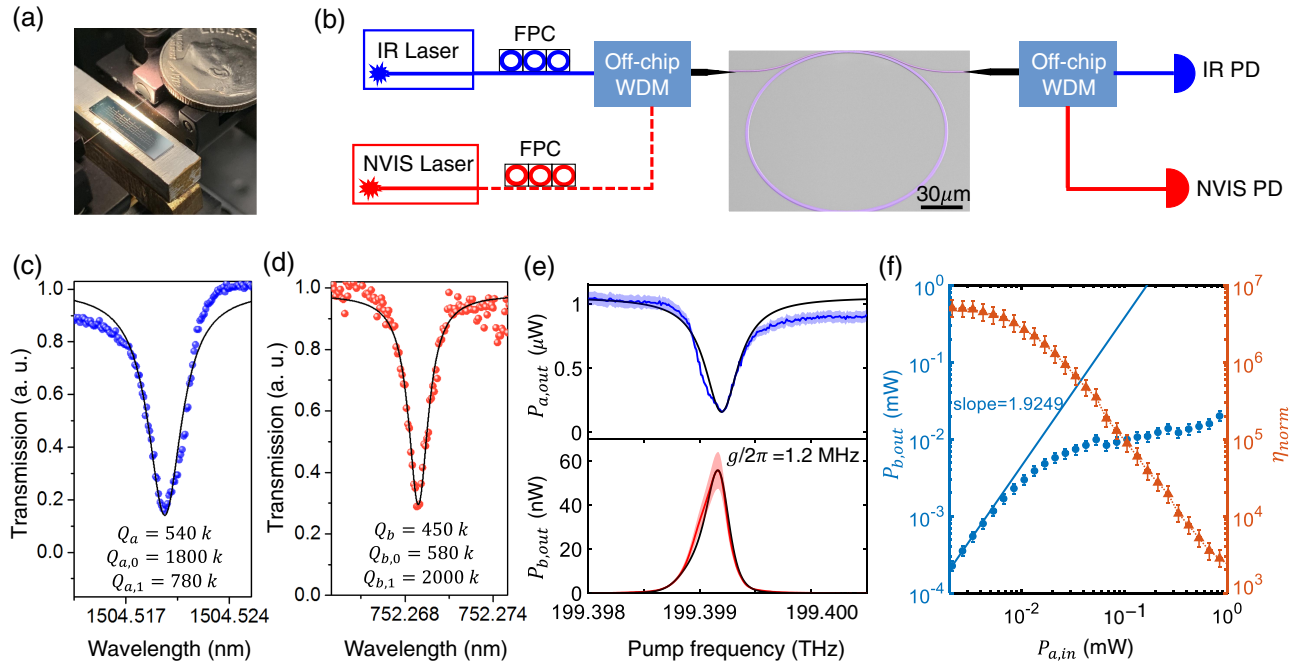


Fig. 3. (a) Photography of the measurement stage with input/output fiber coupling to the PPLNMR chip. The chip sits on an aluminum sample holder affixed to a feedback-controlled heater with a tuning range from 25 to 200°C. (b) Illustration of the experimental setup with a false-color SEM image of the PPLNMR device. IR, infrared; Nvis, near-visible; FPC, fiber polarization controller; WDM, wavelength-division multiplexer; PD, photodetector. (c)–(d) Transmission spectra of the corresponding fundamental TM0 and near-visible TM0 modes for SHG and their extracted Q values. (e) Spectra of the pump resonance and corresponding SH response measured at an optimized temperature. (f) $P_{b,\text{out}}$ - and η_{norm} - $P_{a,\text{in}}$ relations are plotted with the blue dots and orange triangles, respectively. Quadratic relation is fitted (blue solid line) at the low power regime and a normalized conversion efficiency η_{norm} of $5,000,000 \pm 1,200,000\%/W$ is calculated, while the deviation of quadratic dependence and decay of η_{norm} are observed at the high power regime due to the intrinsic photorefractive effect.

$\delta_a = \omega_a - \omega_p$ ($\delta_b = \omega_b - 2\omega_p$) is the detuning for mode a (b). The normalized SHG efficiency is given by $\eta_{\text{norm}} = P_{b,\text{out}}/P_{a,\text{in}}^2$. Based on the measured $\kappa_{a(b)}$ and $\kappa_{a(b),1}$, the coupling rate g could be fitted from the experimental SH response according to Eq. (4).

For the experimental setup, the chip is mounted on an aluminum holder, whose temperature could be globally tuned with an attached resistive heater [Fig. 3(a)]. The devices are optically addressed using lensed optical fibers. As shown in Fig. 3(b), we selectively turn on the telecom and near-visible light sources for the optical Q measurements of the fundamental and SH modes, while only the telecom laser is swept for the SHG measurement. Figure 3(c) presents the measured transmission spectrum of the fundamental mode around 199 THz for the subsequent SHG, which exhibits an intrinsic and loaded Q of 1.8×10^6 and 5.4×10^5 , respectively. Likewise, the intrinsic and loaded Q factors for the corresponding SH mode are measured to be 5.8×10^5 and 4.5×10^5 as indicated in Fig. 3(d). Hence, the intrinsic dissipation rates of the fundamental and SH modes are calculated to be $\kappa_{a,0}/2\pi = \omega_a/4\pi Q_{a,0} = 55.4$ MHz and $\kappa_{b,0}/2\pi = 343.8$ MHz, which correspond to propagation losses of 0.23 dB/cm and 1.44 dB/cm at the telecom and near-visible bands, respectively. Based on the calibrated insertion loss of 8.5 and 10.0 dB/facet for the respective fundamental and SH modes, Fig. 3(e) highlights the measured pump transmission and corresponding SH response at an optimal temperature, showing that a maximum on-chip SHG power of 55.6 nW is obtained with an on-chip pump power of 1.05 μ W. The shaded region corresponds to an on-chip power variation induced by a coupling fluctuation of 5% and 15% for the respective fundamental and SH modes, and thereby an on-chip normalized SHG efficiency η_{norm} is estimated to be 5,000,000%/W with an uncertainty of 1,200,000%/W.

The theoretical pump transmission and SHG output power based on Eqs. (3) and (4) are also plotted with the solid black lines in Fig. 3(e), where a nonlinear coupling strength $g/2\pi$ of 1.2 MHz is consequently fitted. The discrepancy between the measured $g/2\pi$ and the theoretically predicted value of 1.78 MHz is possibly due to nonuniformity inherent to nanofabrication at different azimuthal angles of the microring [28] as well as random duty cycle errors due to the intrinsic defects in the single-crystalline LN thin film [24,29], as implied in the SEM image of the selectively etched PPLNMR mock-up with HF acid [Fig. 2(d)]. Moreover, the power-dependence of the SHG power (blue dots) and the on-chip efficiency (orange triangles) are plotted in Fig. 3(f), where a linear-fitted slope of 1.92 justifies a quadratic dependence of SHG power on the pump power as predicted by Eq. (4) and an η_{norm} of 5,000,000 %/W is confirmed in the low power regime ($P_{a,\text{in}} < 10 \mu\text{W}$). The deviation of the quadratic dependence and degradation of η_{norm} with the increasing pump power are probably attributed to the increasing frequency mismatch between the mode a and b induced by the accumulating photorefractive (PR) effect [30]. Such PR damage remains challenging for a broad class of thin film LN devices, including frequency converters and modulators [24,31,32], and will be investigated carefully in the future. As we are focusing on the nonlinearity at the single-quanta limit, the degraded device performance in the high power regime will not be a limiting factor for our device. The above experimental demonstration justifies the simultaneously optimized $g/2\pi$ of 1.2 MHz and $\kappa_a/2\pi$ of 184.6 MHz via the record-high normalized SHG efficiency and highlights the potential of PPLNMR to play a key role in future quantum photonics applications.

4. DISCUSSION AND OUTLOOK

The present PPLNMR device exhibits a state-of-the-art single-photon anharmonicity FOM of 0.7×10^{-2} in comparison with that of the aluminum nitride (AlN) [33–35], gallium arsenide (GaAs) [36], gallium nitride (GaN) [37], gallium phosphide (GaP) [38], and silicon carbon (SiC) [39] integrated $\chi^{(2)}$ cavities, as indicated in Fig. 4(a). By simply designing the external coupling to the under-coupled condition, the internal FOM ($g/\kappa_{a,0}$) of our device has already reached 0.02. The photonic crystal (PhC) provides another choice for high FOM by taking the advantage over the coupling rate g due to its ultra-small mode volume ($\sim(\lambda/n)^3$), which is around 3 orders of magnitude smaller than that of the typical microrings. However, there are trade-offs in its relatively higher dissipation rate as the device is scaled down and also the difficulty in designing the simultaneous bandgap in fundamental and SH wavebands [40–43].

We note that although the strong coupling condition for photon blockade is not met by the current device, a FOM

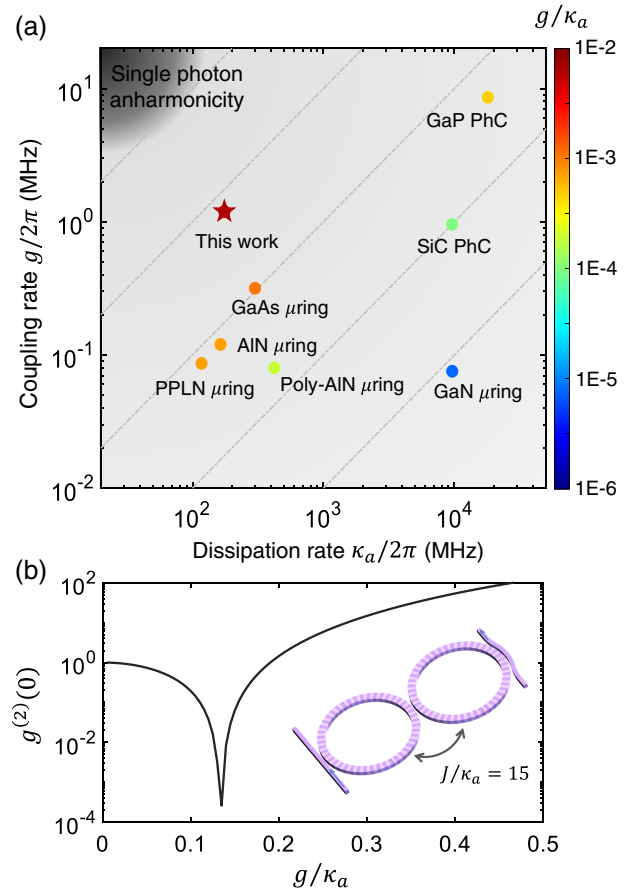


Fig. 4. (a) Coupling rate g and dissipation rate κ_a as well as the corresponding single-photon nonlinear anharmonicity FOM demonstrated in various integrated $\chi^{(2)}$ photonics platforms, including AlN [35], GaAs [36], GaN [37], poly-AlN [33], PPLN [24] μ ring, and GaP [38], SiC [39] PhCs. Note: g values for GaAs, GaN, GaP, and SiC are estimated from the representative data based on the assumptions that both fundamental and SH modes are critical-coupled and $\kappa_b = 2\kappa_a$ when the κ_b is not given. (b) Dependence of the second-order correlation $g^{(2)}(0)$ upon g/κ_a for the fundamental photons in the cavity. The inset depicts a photonic molecule design using two coupled PPLNMRs, allowing a controllable linear coupling strength J between the two fundamental modes. For a J/κ_a of 15, unconventional photon blockade occurs on condition that g/κ_a reaches 10^{-1} .

smaller than unity still promises emitter-free quantum effect in photonic integrated circuits by employing the mechanism of unconventional photon blockade [44–46]. By designing an interferometer in the Fock state space with an ancillary cavity mode, quantum states of deep sub-Poisson statistics have been demonstrated in quantum dot cavity quantum electrodynamics [47] and superconducting resonator [48]. Likewise, we propose that, by introducing another microring resonator and realizing a PPLNMR photonic molecule [49], single photons could be generated from a coherent laser input. The interaction in such PPLNMR photonic molecule can be written as $\hat{H}_{\text{int}}/\hbar = g(\hat{a}_1^{\dagger 2}\hat{b}_1 + \hat{a}_1^2\hat{b}_1^{\dagger}) + g(\hat{a}_2^{\dagger 2}\hat{b}_2 + \hat{a}_2^2\hat{b}_2^{\dagger}) + J(\hat{a}_1^{\dagger}\hat{a}_2 + \hat{a}_2^{\dagger}\hat{a}_1)$, where $\hat{a}_{1,2}$ ($\hat{b}_{1,2}$) represents the respective fundamental (SH) modes in two adjacent microrings. $\hat{a}_1$ and $\hat{a}_2$ are coupled with a linear coupling rate J , which is controlled by the gap between two microrings. The second-order correlation of the intracavity fundamental field is defined as $g^{(2)} = \langle \hat{a}_1^{\dagger}\hat{a}_1^{\dagger}\hat{a}_1\hat{a}_1 \rangle / \langle \hat{a}_1^{\dagger}\hat{a}_1 \rangle^2$, provided that $J/\kappa_a = 15$ (which is achievable and has been demonstrated in Ref. [49]), whose dependence on g/κ_a is numerically investigated and plotted in Fig. 4(b). Accordingly, by further increasing coupling strength g with a reduced microring radius of 40 μm while maintaining low dissipation rate κ_a with a Q of 5 million through an optimized fabrication flow [26,50–52], $g/2\pi$ of 2.35 MHz and $\kappa_a/2\pi$ of 19.2 MHz could be envisioned, which contributes to a FOM of 0.12 and enables the pronounced photon antibunching.

5. CONCLUSION

In conclusion, we have presented the optimization of $\chi^{(2)}$ photon-photon coupling strength toward single-photon nonlinearity in a PPLNMR. Utilizing its largest $\chi^{(2)}$ tensor element d_{33} , and implementing a high-fidelity radial poling with a period of 2.95 μm in an etched z-cut LN microring, a new-record normalized SH conversion efficiency of 5,000,000%/W is demonstrated. Meanwhile, the single-photon coupling rate $g/2\pi$ is estimated to be 1.2 MHz, and thereby a state-of-the-art single-photon anharmonicity FOM of 0.7×10^{-2} is obtained. With 1 order of magnitude improvement, we theoretically propose a PPLNMR photonic molecule device configuration that allows for the remarkable single-photon filtration via unconventional photon blockade effect and paves the way for emitter-free, room-temperature quantum photonic applications, such as quantum light sources, photon-photon quantum gate, and quantum metrology.

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